

Research paper

Energy community incurred grid cost redistribution mechanisms considering different objectives and aspects of fairness

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ABSTRACT

Uniqueness and following individual objectives should be an integral part of energy communities (ECs). However, often-times ECs are regulated quite strictly, thus mitigating their individualism. To provide a certain degree of freedom for ECs, this study introduces three different redistribution mechanisms for EC-incurred grid costs: Income-based redistribution aims at alleviating the financial burden of low-income households, contribution-based redistribution benefits participants providing EC assets (e.g. PV system), while proximity-based redistribution rewards virtual energy flows with short distances from source to sink. The redistribution mechanisms are applied to a fictitious EC of twelve household participants, six being equipped with a rooftop PV system. All three redistribution mechanisms prove to alleviate the financial burden of some EC participants, while EC participation still remains profitable for those participants who are burdened with additional costs due to redistribution. In the case of income-based redistribution the range of additional costs for medium- and high-income households ranges between 5€ and 40€ per year. A similar maximum of additional costs can be reported in case of contribution-based redistribution. Proximity-based redistribution leads to an additional cost range of 30€ to 54€. While fairness is mostly a subjective perception, the redistribution mechanisms can be considered fair in the sense that additional financial burdens in percent are equal for households sharing certain decisive characteristics (e.g. PV system, belonging to medium- or high-income category). The proposed algorithms are not restrained for application to EC-incurred grid costs, and can be applied to other cost components that seem fit for redistribution between EC participants.

Abbreviations

CEC	Citizen energy community
CEP	Clean Energy Package
DSO	Distribution System Operator
EC	Energy Community
EMD	Electricity Market Directive
PV	Photovoltaic
REC	Renewable Energy Community
RED	Renewable Energy Directive

1. Introduction

With legislation for Renewable Energy Communities (RECs) and Citizen Energy Communities (CECs) provided within the recast of the Renewable Energy Directive (REDII) (EUR-Lex, 2018) and the Electricity Market Directive (EMD) (EUR-Lex, 2019) at a European level, EU member states have been obliged to enact national legislation for

energy communities (ECs). Thus, the foundation is laid for citizens to reconquer a certain extent of control of their energy supply, and make an active contribution towards a more sustainable energy future. Since EU guidelines are not precise regarding how certain aspects of ECs should be realised, member states are given a certain extent of freedom regarding the details of transposition into respective national legislations.¹ However, it can be observed that ECs are often-times regulated quite strictly, which limits their potential to freely follow individual objectives. This study's goal is to provide ECs with a certain degree of freedom to follow their unique objectives and values.

For energy exchanges within the EC (between peers), certain costs arise. The two most significant parts of EC-incurred costs are on the one hand (i) costs for energy purchase from community peers,² and on the other hand (ii) costs for grid usage. While the price for inner-community energy exchanges is usually decided by the EC participants collectively, grid charges cannot be decided by the community, but are simply imposed by the DSOs — being regulated monopolies, having to recover their expenses, e.g. from infrastructure maintenance and

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¹ Some additional impressions of how ECs are managed and operated throughout different European countries is provided in Appendix A.

² Certainly, according revenues are accounted for selling energy to peers.

grid operation. Although an EC cannot alter the height of grid charges (and/or their allocation to individual participants by the DSO), they still have the option to apply redistribution mechanisms ex-post.³ This is one significant degree of freedom to shape cost distribution such that it fits to an EC's unique characteristics and objectives.

Therefore, in this study, three different redistribution mechanisms for EC-incurred grid costs⁴ are developed and investigated. The first mechanism suggests a redistribution of EC-incurred grid costs based on households' income levels, the second mechanism is based on the contribution of participants to the EC, while the third mechanism takes the proximity between EC members into consideration. It needs to be emphasised that the proposed redistribution mechanisms of EC-incurred grid costs are independent from whichever grid tariff is imposed by the DSO to its grid users. The redistribution takes place ex-post and thus is neither implemented by the DSO, nor is the DSO directly affected by it. It simply is the choice of an EC to realise their subjective perception of distributional fairness⁵ with regard to EC-incurred grid costs⁶. All three redistribution mechanisms are applied to a fictitious EC consisting of twelve single-family households, out of which six are equipped with a rooftop PV system. The impact of the proposed redistribution mechanisms on the individual EC participants shall be evaluated, and aspects of fairness discussed.

The remainder of this paper is structured as follows. Section 2 provides an overview of the current state-of-the-art in scientific literature along with the contribution this paper aims to make. The developed redistribution mechanisms, the selected case study, and relevant data is introduced in Section 3. Section 4 shows results regarding the impact of redistributing EC-incurred grid costs based on different criteria, while Section 5 discusses aspects of fairness in this context. Conclusions are derived in Section 6.

2. State-of-the-art

With respect to this study's focus, the selection of literature provided in this section encompasses studies focusing on grid tariff fairness in Section 2.1, profitability aspects of ECs in general in Section 2.2, before paying special attention to cost and revenue distribution within ECs in Section 2.2.1. Section 2.3 discusses literature regarding fairness aspects in the context of ECs.

2.1. Grid tariff design and fairness

Within recent years, a variety of different grid tariffs have been designed and assessed in scientific literature. According to Ansarin et al. (2022) seven residential grid tariff designs are distinguished: flat-rate, increasing-block pricing, Ramsey pricing, real-time pricing/dynamic pricing, time-of-use, peak-pricing, and demand charges. Tariff design is usually not done randomly, but follows a number of principles, such as sustainability/sufficiency of revenue, equity/non-discriminatory access,

³ This means that the DSO is not affected by the application of the ex-post redistribution mechanisms at all.

⁴ The term 'EC-incurred grid costs' means costs that are incurred by using the distribution grid for energy exchanges between EC participants. In this work, only the energy-dependent cost components (in €/kWh) are considered, while power-dependent cost components (€/kW) are neglected.

⁵ Distributional fairness in general means the fair distribution of resources. In this work, the resources subject to redistribution for the sake of distributional fairness are EC-incurred grid costs. Fairness is perceived subjectively by different people, depending on their personal values and backgrounds. This is why this study will introduce three entirely different redistribution mechanisms in order to cover a certain spectrum of distributional fairness.

⁶ It needs to be mentioned that such redistribution mechanisms economically benefit some, while disadvantaging others. However, it should be ensured that EC participation per se still remains profitable for all participants, despite the application of redistribution mechanisms.

economic efficiency, transparency, stability, consistency with the larger regulatory framework, and additivity of cost elements (Reneses and Ortega, 2014). Vaughan et al. (2023) add the criteria of privacy and implementability to this list. However, grid tariffs and their optimal design are usually developed continuously, taking into account current needs and challenges. In this respect, a systematic review of the different challenges posed by an increasing amount of distributed generation is conducted by Picciariello et al. (2015). Going in a similar direction, Schittekatte et al. (2018) show that new technology choices for grid users (such as rooftop PV systems and battery storages) can evoke efficiency and equity issues in case distribution grid charges are ill-designed, while Ansarin et al. (2022) indicate that accumulating distributed generation for self-consumption greatly reduces inefficiencies. In order to be able to properly assess the impact of different grid tariffs, a feasibility evaluation method to review dynamic distribution tariffs is introduced in Christensen et al. (2021) based on technical, economic, social and regulatory dimensions; The time-of-use tariff is found to be the most feasible. However, in order to prevent that time-of-use tariffs disadvantage low-income households, a flexible, fairness-based time-of-use tariff design is proposed by Li et al. (2022b). In light of the difficult task to choose appropriate grid tariffs in an environment with increasing shares of distributed energy resources, Hennig et al. (2022) design indicators for assessing the performance of distribution network tariffs. Continuing this effort, Heider et al. (2024) contribute by introducing a process to identify and weigh relevant decision criteria for network tariff design, and translate these into quantitative performance indicators.

While the above mentioned criteria for grid tariff design seem logical and comprehensible, the individual grid tariffs themselves are heatedly discussed with respect to fairness. The fairness-principle proposed by Ansarin et al. (2022) suggests that equal customers should pay equally for the same product. While this might sound straightforward, Khan et al. (2023) highlight the complexity of this matter by finding that the design of equitable tariffs requires temporal and spacial granularity in tariff structures. However, 'fairness' is to a large extent a subjective perception, wherefore (Trong and Yang, 2023) investigates fairness perception for grid tariff models based on a survey of Danish households. Neuteleers et al. (2017) also assess fairness perceptions of different grid tariffs, finding that transport and capacity charges are evaluated positively, Ramsey pricing most negatively. Flat rates and peak pricing are situated in the middle. Adding another point of view, Winther and Sundet (2023) point out that justice perspectives related to the introduction of a new tariff are perceived differently by grid companies and household customers. Also, grid tariffs and their perceived fairness are closely related to the concept of distributional justice. To divide common costs, Neuteleers et al. (2017) distinguish between (i) equal allocation, (ii) allocation based on the ability to pay, (iii) allocation based on costs, (iv) allocation based on benefits derived from the good, and (v) allocation based on desire. Also with a focus on distributional justice, Bergaentzle et al. (2022) combine tariff designs and redistributive mechanisms to strike a balance between time-differentiated signals, revenue stability for the utility, limited grid costs for vulnerable households, and the promotion of electrification. Finally, since grid tariffs and distribution grid design are mutually dependent, the latter also follows a number of principles, researched in Meeus (2018).

2.2. Profitability aspects of ECs

Despite financial gain may not be the main purpose of ECs as per the European Directives ('...the primary purpose of which is to provide environmental, economic or social community benefits for its shareholders or members or for the local areas where it operates, rather than financial profits' RED II, Article 2 (16c)), a certain economic viability is inevitable for large parts of the population. Thus, a number of studies have addressed profitability aspects from manifold points of view. Petrichenko

et al. (2022) compare profitability of being a ‘distributed prosumer’ or an ‘EC prosumer’,⁷ finding that participating in an EC leads to increased cost-savings (approx. 20% higher) in comparison to acting as an individual prosumer. The finding that establishing ECs is profitable is also confirmed by results provided in Felice et al. (2022). Going one step further, profitability of establishing ECs in individual buildings or across building borders is investigated in Fina et al. (2019) for different settlement patterns by introducing a MILP⁸ optimisation model. Thereby, it is found that establishing an EC across building borders is most profitable in rural areas. Rocha et al. (2023) propose a three-stage model, firstly optimising schedules of individuals, secondly optimising across the community and thirdly solving local grid constraints through redispatch. Thereby it is always guaranteed that individual community members do not suffer financial losses. Mehta and Tiefenbeck (2022) perform a deterministic techno-economic analysis to learn about the impact of EC configurations on the profitability and find that an EC’s size is irrelevant for prosumer profitability, as well as for community self-sufficiency. Further research in the context of community set-up is provided by Volpato et al. (2022) by introducing general guidelines for the economically optimal aggregation of prosumers in ECs. Authors find that in case of optimal synergies between prosumers, costs can be reduced in the range of 15%–20%. Besides the EC set-up itself, it is found in Gjorgievski et al. (2023) that regulated charges and the solar potential have the greatest influence on cost savings. The latter is confirmed by findings of Fina et al. (2021). Also a decisive factor for profitability is an EC’s assets along with the means of finance. Optimal investments into PV and storages within ECs are elaborated on in Radl et al. (2020), making use of a linear optimisation model. It is found that under current market conditions, storages are rarely profitable for increasing PV self-consumption; however, potential can be detected in case power pricing would be applied. Since investments in renewable generation units can pose a significant financial burden for ordinary citizens, Monsberger et al. (2021) prove with the help of a mixed-integer linear optimisation model that energy supply contracting helps citizens to overcome the barrier of high upfront investment costs.

In order to achieve the best possible outcome regarding profitability, the optimal planning and set-up of ECs in terms of participant composition and energy technology mix is crucial. Therefore, aspects of optimal community design are addressed by a number of studies, such as Sousa et al. (2023), Lazzari et al. (2023) and Braeuer et al. (2022). With respect to this study’s grid focus, it is also important to highlight that the composition of the EC can be decisive of a community’s grid-friendliness, an issue being addressed in Sudhoff et al. (2022), also in the context of economic benefits.

Also with respect to profitability aspects, business models are of interest: Iazzolino et al. (2022) illustrate a theoretical comparison of business models for ECs, Karami and Madlener (2022) focus in particular on business models for peer-to-peer trading, and Barnes et al. (2024) explore value creation pathways in the business model context. Mihailova et al. (2022) start to broaden the perspective towards exploring sustainable value co-creation between citizens participating in ECs and other stakeholders, while (Kubli and Puranik, 2023) elaborate on a typology of business models for ECs considering current and future design options. Finally, Fina and Monsberger (2023) entirely focus on business model approaches for stakeholders in the context of ECs.

To conclude the discussions on profitability of EC participation and establish a link to grid tariffs and their design, it is important to emphasise that grid tariff designs can have significant impacts on the profitability of EC participation. Especially in case of designs such as peak pricing or time-of-use tariffs, participating in an EC could be a

significant advantage since synergy effects can be used to lower power peaks, and/or demand in general when tariffs are highest. Through participating in an EC, individuals are in favour of a wider range of possibilities to adapt to novel grid tariff concepts, since the collective usually has more room for manoeuvre compared to an individual.

2.2.1. Cost and revenue distribution

Li et al. (2021a) focus, among others, on a review of cost allocation methods (e.g. flat energy pricing, base and peak, Ramsey pricing etc.), and their applicability to ‘integrated community energy systems’ (ICESs). However, since there is no universal cost allocation method, the tariff structure is not definite in ICESs. Nevertheless, it is important to provide sufficient information to the end-users such that they can fully understand how their energy consumption is charged. This is the basis for social acceptance of cost allocation in ICESs, an issue that is further investigated by the same authors in Li et al. (2021b). Multi-group and multi-criteria decision-making methods are applied to support the selection of a cost allocation method due to the participation of different community members with various preferences. Furthermore, Li et al. (2022a) compute the energy bills for community members in an ICES for various cost allocation methods and determine their cost reflectiveness. Extending research in this direction, they further investigate a selection of cost allocation methods (flat energy pricing, time-of-use and segmented energy pricing) with respect to two different investment models (third-party financing and self-financing) (Li and Okur, 2023). Grzanic et al. (2021) propose a novel centralised post-process sharing method by introducing a two-stage mechanism. If in the first stage it is discovered that some members face increased cost, the second stage is used to distribute savings such that all members benefit. Also working with a two-stage approach, Mustika et al. (2022) introduce a two-stage strategy for optimal operation and billing in an EC. In order to share the community costs in a fair manner, two different cost distributions are proposed in Hupez et al. (2021). The first one adopts a distribution key based on the Shapley value, while the other relies on a natural consensus defined by a Nash equilibrium. Also using a game theoretic approach, profit sharing in ECs is addressed through an asymmetric Nash bargaining model by authors in Wu et al. (2022). Roberts et al. (2022) focus on efficient, effective and fair allocation of costs and benefits in residential ECs, taking into account a range of tariff structures. The importance of this section shall finally be substantiated by findings of Siqin et al. (2022), highlighting that the enthusiasm of cooperating within the community can be enhanced by comprehensive profit (re-)allocation.

2.3. Fairness aspects in ECs

With respect to ECs, fairness aspects are also subject to individual perceptions. For example, Moret and Pinson (2019) define fairness such that a system is entirely fair in case the traded energy volumes are equal for all participants. A fair energy sharing method for collective self-consumption is proposed by Gjorgievski et al. (2022), together with a framework to study the actual fairness of the proposed method. In Oh (2022) an operation strategy is proposed for communities based on PV with a special focus on resource fairness to achieve a fair distribution of electricity bill savings. Also focusing on fairness from a monetary point of view, Casalicchio et al. (2021) are concerned to determine an optimal allocation method for a fair distribution of benefits in an EC, subject to participants’ own contribution to the system. Results generated by Oh and Son (2020) show that social welfare is maximised by matching producer and consumer pairs which have similar characteristics. Thereby, profit fairness is being achieved through balanced transaction prices. Talking about specific prosumer and consumer pairs, it needs to be mentioned that in some countries (e.g. Austria) ECs are realised such that the DSO is obliged to allocate electricity based on a static or dynamic distribution key to the individual EC participants. This means that EC participants only know the amount of electricity

⁷ A ‘distributed prosumer’ is considered an end-user with their own rooftop PV system for self-consumption. An ‘EC prosumer’ indicates a prosumer (entity with a PV system) participating in an EC.

⁸ MILP = Mixed-Integer Linear Programming.

received from or sold to the EC, but it is not known from which peer exactly energy is received, or to who exactly energy is sold (in other words, peer-to-peer information is missing). To address this issue, a method for ex-post electricity allocation based on participant preferences is developed in [Fina \(2023\)](#). This then allows the realisation of a certain kind of subjective fairness by being able to charge individual prices for energy exchanges instead of having to stick with one price per community. A fair ex-post cost allocation is studied by [Yang et al. \(2021\)](#) based on nucleolus which addresses fairness by minimising the dissatisfaction of all the players. It is assumed that a player is satisfied or dissatisfied based on the allocated costs, meaning that lower costs lead to increased satisfaction of profit-oriented players. [Limmer \(2023\)](#) indicate that benefit distribution schemes such as the nucleolus, Shapley value and Shapley-core are known to perform well in terms of fairness and stability, but full fairness and stability at the same time cannot be guaranteed. It is found that the community size can have a significant impact on both.

Since ECs are meant to have social benefits through empowering vulnerable citizens and help fight energy poverty, studies are increasingly concerned with ECs' social impact ([Bielig et al., 2022](#)), which plays a significant role in the fairness discussion. However, to achieve social benefits, citizen participation is required, a topic being addressed in [Wahlund and Palm \(2022\)](#), emphasising that energy citizenship is related to behavioural change of individuals. In order to achieve a broad engagement of citizens, acceptance of EC initiatives is vital ([Hogan et al., 2022](#)), and also the mitigation of energy poverty needs to receive increased attention in this context ([Bode, 2022](#)).

2.4. Progress beyond the current state-of-the-art

The literature review presented in Section 2.1 shows the existence of multiple grid tariff design options, which are perceived and evaluated differently from the grid users with respect to fairness. However, grid users are not entitled to choose their grid tariff (which is usually rather given by the DSO), wherefore their individually subjective perception of fairness.⁹ is not reflected in their grid charges. When it comes to ECs, a solution to this limitation of personal choice can be provided by introducing ex-post grid cost redistribution mechanisms for EC-incurred grid costs. In this way, EC participants are given an opportunity to approach the matter of distributional fairness with regard to grid charges uniquely, a matter that has not been addressed in scientific literature up to this point.

Since ECs are concepts carrying individuality in their DNA, and incorporating fairness aspects from different angles is vital as literature emphasises in Section 2.3, this study aims at providing an additional degree of freedom for ECs by developing three redistribution mechanisms for EC-incurred grid costs, ensuring that economic viability is still preserved for all participants.¹⁰ The proposed EC-incurred grid cost redistribution mechanisms are the following:

- Income-based redistribution to alleviate the financial burden of low-income households.
- Contribution-based redistribution to reward participants' previous
- Proximity-based redistribution to account for energy consumption as local as possible.

The need for novel redistribution mechanisms of certain cost/revenue components can further be manifested by the literature review provided in Section 2.2.1: A number of studies concerned with cost/revenue distribution in ECs focus on specific tariffs (e.g. flat energy pricing,

time-of-use, etc.) and their implications on different participants' financials. However, such tariffs would still be imposed by a DSO, and thus would not reflect personal choice. Other works propose multi-stage cost/revenue allocation processes, having a slight disadvantage with regard to increased complexity in practical implementation and actual comprehensibility for EC participants. The third group of studies use game theoretical approaches to address cost/revenue distribution issues, which is of significant scientific importance, but lacks practical applicability. This stresses the need for new redistribution mechanisms that are able to reflect personal choice and individual perceptions of distributional fairness, while having the potential to be actually implemented in practice and understood by EC participants. Moreover, never have (re-)distribution mechanisms been developed in a similar manner as proposed in this work, nor have any (re-)distribution mechanisms been introduced specifically for redistributing EC-incurred grid costs.

The developed redistribution mechanisms are applied to a fictitious EC in order to explore the impacts upon different EC participants. Matters of fairness are discussed based on the achieved results in order to develop a sense of practical applicability.

3. Method, model and data

The EC used as a case study is introduced in Section 3.1, cost calculations and according assumptions are lined out in Section 3.2, before the mathematical formulation of the redistribution mechanisms is provided in Section 3.3.

3.1. Case study

A fictitious EC with twelve household participants is used as a case study. Six of these households are equipped with rooftop PV systems of different capacity and orientation (south, east and west). Irradiation profiles are used with a 15-minute resolution. In order to make the case study as realistic as possible, all households are allocated different, real-measured load profiles at 15-minute resolution, in order to guarantee for realistic variance. The EC is assumed to be located in Vienna, Austria, with the respective solar irradiation. With respect to the introduction of different redistribution mechanisms for EC-incurred grid costs, the participants are further distinguished by annual income (low, medium, high) and geographical proximity (two proximity areas based on the assumed connection to the distribution grid). These different participant characteristics are visualised in [Fig. 1](#).

3.2. Cost calculations, prices and tariffs

Prices and tariffs for energy purchase from or sales to (i) the conventional supplier or (ii) the EC are provided in [Table 1](#). Costs for electricity usually consist of the energy price, grid tariff components, and taxes & levies. For this study, the default price of purchasing electricity from a conventional supplier is a real-world price of the Austrian supply company Wien Energie ([Wien Energie, 2023](#)). The energy price for purchasing and selling energy within the EC is chosen such that it is profitable to purchase from the EC (since the price for energy purchase within the EC is smaller compared to the one the supplier offers) as well as to sell to the EC peers (since revenues are higher for selling to the EC than to the conventional supplier).¹¹ It needs to be mentioned that costs considered in this study do only concern the energy-dependent components (costs per kWh), and do not account

⁹ The concept of fairness being perceived differently by individuals is underpinned further in [Appendix B](#).

¹⁰ Literature of Section 2.2 shows the importance of profitability, despite being not the main objective of EC participation.

¹¹ Choosing the energy price in this way ensures economic viability for all EC participants (prosumers as well as sole consumers). However, it is not binding for ECs to follow this pricing logic; energy can also be gifted, or participants can also accept energy prices that lead to a worsening of their financial situation, e.g. for the sake of benefiting others. In this study, however, the pricing logic of choosing the energy price in a way that every participant benefits economically is applied.

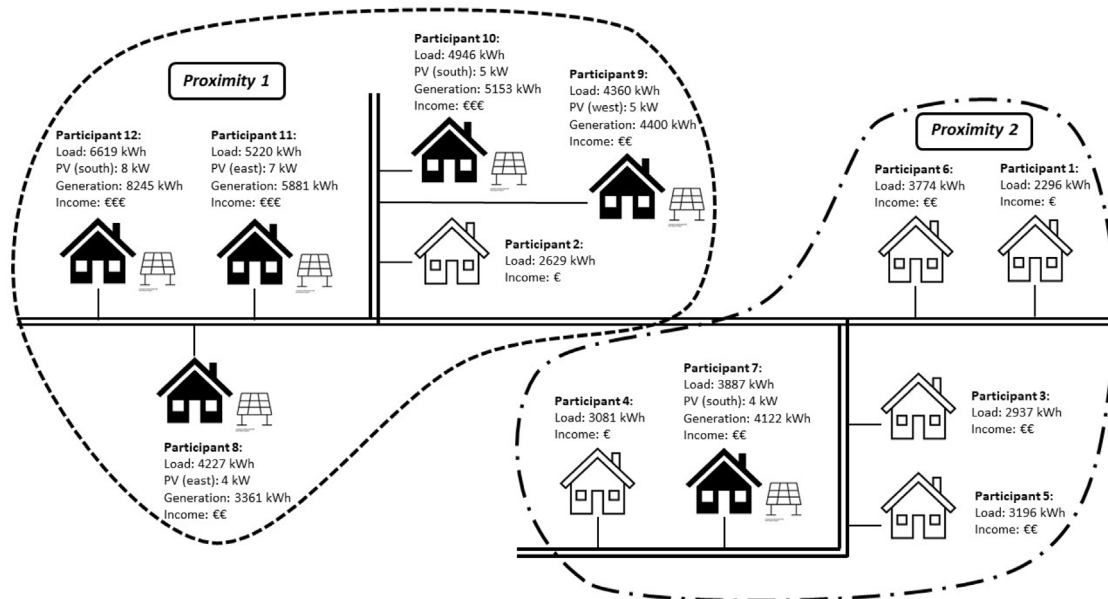


Fig. 1. Fictitious energy community with twelve participants distinguished per annual load, generation (PV of different size and orientation or no PV), income (low – €; medium – €€; high – €€€) and proximity (proximity area 1 comprising households 2, 8, 9, 10, 11, 12 and proximity area 2 comprising households 1, 3, 4, 5, 6, 7).

Table 1
Electricity price composition; * component including VAT (20%) and usage fee (6%); ** component including VAT only. Price components are derived from [Wien Energie \(2023\)](#), [Wiener Netze \(2023\)](#) and [Rechtsinformationssystem des Bundes \(2023b\)](#).

Electricity price for purchase from the supplier		
Component	net value	gross value
Energy price	33.2612c/kWh	42.3082c/kWh *
Grid usage fee	5c/kWh	6.36c/kWh *
Grid loss surcharge	0.593c/kWh	0.754c/kWh *
Electricity surcharge	0.1c/kWh	0.12c/kWh **
Total costs		49.5425c/kWh
Electricity price for purchase from the EC		
Component	net value	gross value
Energy price	22c/kWh	- (exempt f. taxes)
Grid usage fee	5c/kWh	6.36c/kWh *
Grid loss surcharge	0.593c/kWh	0.754c/kWh *
Electricity surcharge	0.1c/kWh	0.12c/kWh **
Total costs		29.234c/kWh
Price for energy sales to the supplier		
Component	net value	gross value
Energy price	15c/kWh	-
Grid loss surcharge	2.489c/kWh	-
Total revenues		12.511c/kWh
Price for energy sales to the EC		
Component	net value	gross value
Energy price	22c/kWh	-
Grid loss surcharge	2.489c/kWh	-
Total revenues		19.511c/kWh

for the power-based components. Regarding grid costs it is assumed that full grid charges apply to EC participants. This means that EC participants are not being advantaged compared to other grid users (not being part of an EC), since reduced grid charges would only be suitable if ECs would act in a grid-friendly manner, which is not the default case.

Since this study is about redistributing EC-incurred grid costs and investigating the respective financial impact on the participating households, a short overview of how costs are calculated is given in the following.

In the default situation (assuming that no EC is established), and thus each household would remain on its own, total annual costs of electricity would be calculated as follows:

- total load of households without a PV system and residual load (after self-consumption) of households with a PV system would be covered by the conventional supplier, wherefore energy costs and grid costs are charged.
- for households owning a PV system, costs would be reduced by revenues generated by selling the surplus generation to the conventional supplier for a certain feed-in-tariff (minus charges for grid losses that are also applicable to generated amounts of energy fed into the grid).

In case an EC is installed, smaller amounts of energy are purchased from and/or sold to the conventional supplier. This is due to energy exchanges between community peers. Therefore, items in the annual cost calculation are different to the default situation (with no EC installed):

- Total costs are composed of:
 - costs for purchasing energy from community peers (at a lower price compared to the conventional supplier, please compare to cost items given in [Table 1](#))
 - charges for using the grid for energy purchase from community peers
 - costs for purchasing energy from the conventional supplier to cover the residual load (which is not covered either by self-consumption or by purchasing from the EC)
 - charges for using the grid for energy purchase from the conventional supplier
- Total revenues are composed of:
 - revenues for selling energy within the EC
 - revenues for selling the surplus generation to the conventional supplier
 - both listed revenue items are reduced by the grid loss surcharge that is applicable also to sold amounts of energy

The EC-incurred grid charges (to which the redistribution mechanisms are applied) per household are thus composed of

- charges for using the grid for energy purchase from community peers
- grid loss surcharges for selling energy to the EC

3.3. EC-incurred grid cost redistribution mechanisms

The main objective of this study is to develop algorithms that allow for the redistribution of EC-incurred grid costs¹² based on different criteria, such as:

- Participants' financial situation with regard to annual income (Section 3.3.1)
- Contribution in terms of assets beneficial to the whole community (Section 3.3.2)
- Geographical location of participants in the distribution grid and thus proximity towards each other (Section 3.3.3)

Literature presented in Section 2.3 clearly shows that fairness aspects are as diverse as ECs themselves. Therefore, when it comes to choosing fairness-criteria on which this work's EC-incurred grid cost redistribution mechanisms are based, it is impossible to choose them such that every possible fairness-angle is covered. However, the decision to focus on (i) income-based, (ii) contribution-based, and (iii) proximity-based redistribution in this study can be justified as follows. Income-based redistribution (reducing costs for low-income households) is an attempt to deliver on the REDII, specifying that RECs should help to fight energy poverty (REDII, (67)), and that participation in RECs should be accessible to all consumers, including those in low-income or vulnerable households (REDII, Article 22, (4f)) (EUR-Lex, 2018). Contribution-based redistribution (reducing costs for PV system owners) can be justified by the need to further increase the shares of decentralised renewable generation units and aims to provide an additional incentive to invest in PV and participate in an EC¹³ (Cox et al., 2016). Despite being practically the opposite to income-based redistribution (because the more wealthy participants are privileged in the contribution-based approach) it is still valid to grant PV system owners – who had to conduct investments without which the establishment of ECs in the household sector would hardly be possible – additional financial benefits. The approach of basing allocation or redistribution mechanisms on participant's own contribution to the system is also validated by Casalicchio et al. (2021). Lastly, the decision for proximity-based redistribution (mitigating grid costs for energy exchanges in immediate proximity) is made with respect to the fact that consuming electricity close to its source is efficient as well as beneficial for stable grid operation. In this context it needs to be emphasised that the third option of proximity-based redistribution is the only one which is certainly compliant with the needs of DSOs. In this respect, a clear line of this study's goals can be drawn: The redistribution mechanisms are not intended to be beneficial to the DSO, or incentivise grid-friendly behaviour of ECs. The redistribution mechanisms are designed for ECs to realise their subjective perception of fairness in the distribution of EC-incurred grid costs ex-post, independent of any grid tariff design and the therewith associated costs imposed by the DSO.

However, before redistribution mechanisms can be applied, EC-incurred grid costs need to be determined. This means that knowledge

of how and how much energy is distributed within the EC is required. A simulation model¹⁴ determines the exact amounts of energy that are (i) self-consumed by the participants with a PV system, and (ii) shared within the EC by using a dynamic distribution key.¹⁵ Therefrom, also amounts of energy which need to be additionally purchased from the conventional supplier to cover the residual load can be derived, as well as the surplus generation sold to a conventional supplier due to not being used by self-consumption or within the EC. With the amounts of energy known, according costs can be calculated. Out of the different cost-items, the amount of grid costs incurred by inner-community energy exchanges are the ones relevant for the redistribution mechanisms.

For the presentation of the methodology it needs to be mentioned that the individual calculation steps are shown in detail, and the parameter names are deliberately chosen rather long, but with the intention that the names of the parameters speak for themselves, enhancing comprehensibility. Parameters can be time-dependent (t) and/or household-dependent (h). The total number of time-steps is $T = 35040$ (quarter-hours per year), the total number of households is $HH=12$.

3.3.1. Redistribution based on financial situation

Making the financial situation the decisive factor of EC-incurred grid cost redistribution can be considered an attempt to satisfy the requests of the REDII (EUR-Lex, 2018), namely to fight energy poverty (REDII, (67)) and provide EC access to all customers, including those in low-income or vulnerable households (REDII, Article 22, (4f)). Reducing or, in this case, extinguishing EC-incurred grid costs of participants with limited financial means has the potential to alleviate their financial burden. However, this implies that other EC participants need to accept higher costs, since, after all, the total amount of grid expenses needs to be covered.

In this work, three types of income-levels are distinguished¹⁶:

- low-income households (€): Participants 1, 2, 4
- medium-income households (€€): Participants 3, 5, 6, 7, 8, 9
- high-income households (€€€): Participants 10, 11, 12

The redistribution mechanism is set up such that the total EC-incurred grid costs of low-income participants are being borne by medium- and high-income households. The details of the algorithm are explained in the following, with the according nomenclature being provided in Table 2.

The redistribution mechanism needs information to which income category a household belongs, which is supplied via an array $I = [0, 0, 1, 0, 1, 1, 1, 1, 1, 2, 2, 2]$. Based thereon, the total amount of EC-incurred grid costs of low-income households needing to be redistributed (C_{G2S}) can be determined along with the savings that medium- and high-income households obtain by participating in the EC ($C_{d_m}[h]$,

¹⁴ The simulation model has been developed and introduced in the course of Fina et al. (2022), wherefore I would like to refer to this previous publication for further details.

¹⁵ The (PV) generation available within the EC can be allocated to the individual participants using either a static or dynamic allocation key. Static allocation follows the principle that each participant is entitled to a fixed share of the available generation at all time-steps. Dynamic allocation, in contrast, follows the principle that participants with a higher demand at a certain time-step are entitled to higher shares of the available generation. This means that the allocation shares in case of dynamic allocation vary per time-step, depending on the participants' individual demand. Since dynamic allocation is thus the more efficient option, this study assumes dynamic allocation to be applied for energy allocation within the EC. More information regarding static and dynamic allocation can be extracted from Fina et al. (2022).

¹⁶ It is out of this study's scope to assign actual monetary values to the assumed income levels, since this can be different for each country or region. Moreover, actual monetary values are not relevant to the redistribution mechanism.

¹² EC-incurred grid costs are costs arising due to using the distribution grid for (virtual) energy flows between community peers. In this study, only energy-related costs (costs per kWh) are considered, while power-based costs are left aside. However, it needs to be noted that applicability of the developed algorithms is not restricted and can be applied to any kind of cost term.

¹³ Distributed generation is considered an integral part of a sustainable energy future (Nadeem et al., 2023) due to its potential to support development and climate objectives.

Table 2
Nomenclature for income-based redistribution.

Parameter	Dimension	Explanation
C_d	[HH]	annual savings through EC participation per HH
$C_{d,h}$	[HH]	annual savings through EC participation for high-income HHs
$C_{d,m}$	[HH]	annual savings through EC participation for medium-income HHs
C_{G2S}	[1]	total EC-incurred grid costs of low-income HHs to be distributed among other HHs
$C_{G2S,h}$	[1]	total amount of additional costs for all HHs of the high-income category
$C_{G2S,h,HH}$	[HH]	amount of additional costs for each high-income HH
$C_{G2S,m}$	[1]	total amount of additional costs for all HHs of the medium-income category
$C_{G2S,m,HH}$	[HH]	amount of additional costs for each medium-income HH
$C_{G,lot}$	[HH]	annual grid costs within the EC per HH
I	[HH]	array indicating to which income category a HH belongs
S_h	[HH]	high-income HHs' share on savings of all high-income HHs in total
S_m	[HH]	medium-income HHs' share on savings of all medium-income HHs in total
s_h	[1]	share of EC-incurred grid costs that HHs of the high-income category overtake in total
s_m	[1]	share of EC-incurred grid costs that HHs of the medium-income category overtake in total

Table 3
Nomenclature for contribution-based redistribution.

Parameter	Dimension	Explanation
$C_{G,EC,HHwPV}$	[HH]	annually EC-incurred grid costs per HH with a PV system
$C_{G,lot}$	[HH]	annually EC-incurred grid costs per HH
$C_{G2S,HHnPV}$	[HH]	additional costs borne by HHs without PV
Con	[HH]	indication of HHs' contribution to the EC (having a PV system or not)
LC	[T, HH]	load covered by the EC per HH and time-step
LC_{HHnPV}	[T, HH]	load covered by the EC only for households without a PV installed (contribution = 0)
$s_{LC,HHnPV}$	[HH]	share of the load covered within the EC only of HHs without a PV system

$C_{d,h}[h]$), see Eq. (1). The latter will be decisive for the share of additional costs to be borne by medium- and high-income households.

for h in range(HH):

if $I[h] == 0$:

$$C_{G2S} = C_{G2S} + C_{G,lot}[h]$$

else if $I[h] == 1$:

$$C_{d,m}[h] = C_d[h]$$

else:

$$C_{d,h}[h] = C_d[h]$$

end

Based on the savings each medium- and high-income HH can make due to EC participation, their share on the total savings per income-category can be determined as shown in Eq. (2).

$$S_m[h] = \frac{C_{d,m}[h]}{\sum_{h \in HH} C_{d,m}[h]}$$

$$S_h[h] = \frac{C_{d,h}[h]}{\sum_{h \in HH} C_{d,h}[h]}$$
(2)

Since this study distinguishes three income categories, it is assumed that medium-income HH are willing to overtake one-third of the EC-incurred grid costs of low-income households ($s_m = 1/3$) and high-income HH agree to additionally cover the remaining two-thirds ($s_h = 2/3$). This allows for determining the total amount of costs that needs to be borne by households belonging either to the medium- or high-income category (Eq. (3)).

$$C_{G2S,m} = C_{G2S} \cdot s_m$$

$$C_{G2S,h} = C_{G2S} \cdot s_h$$
(3)

In a next step, additional costs incurred for the medium- and high-income category need to be further distributed to individual households (see Eq. (4)).

$$C_{G2S,m,HH}[h] = C_{G2S,m} \cdot S_m[h]$$

$$C_{G2S,h,HH}[h] = C_{G2S,h} \cdot S_h[h]$$
(4)

Lastly, the EC-incurred grid costs of all households needs to be newly determined, meaning that the original EC-incurred grid costs (before redistribution) of...

- low-income households are set to zero
- medium-income households increase by $C_{G2S,m,HH}[h]$
- high-income households increase by $C_{G2S,h,HH}[h]$

3.3.2. Redistribution based on contribution to the community

The second approach proposed in this study is about redistributing EC-incurred grid costs such that participants with a higher contribution to the EC benefit more than others. 'Contribution' is manifested in form of assets that are essential for the operation of the EC, e.g. PV systems. The reasoning behind this is that without the presence of generation units, an EC could not be realised at all. To reward such contribution, EC-incurred grid costs of participants with a PV system are redistributed to be borne by community members without a PV system.

The contribution-based redistribution algorithm (nomenclature provided in Table 3) needs information, which households are equipped with a PV system, and which are not. This is done in form of an array, filled with zeroes (no PV system) and ones (PV system available) per household: $Con = [0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1]$.

The redistribution algorithm starts to extract the load of households without PV system that has been covered by energy from the EC¹⁷ (Eq. (5)). The amounts of these households' load covered by the EC (which is also an indicator how beneficial the EC is to them) is decisive for the redistribution of EC-incurred grid costs. Also, Eq. (5) extracts the grid costs of all households equipped with a PV system to have these amounts ready for redistribution.

for h in range(HH):

if $Con[h] == 0$:

$$LC_{HHnPV}[:, h] = LC[:, h]$$

else if $Con[h] == 1$:

$$C_{G,EC,HHwPV}[h] = C_{G,lot}[h]$$

end

In a next step, the "share of benefit" (based on load covered within the EC) of non-contributing-households is determined as shown in

¹⁷ These values are obtained previously by allocating amounts of energy based on a dynamic distribution key.

Table 4
Nomenclature for proximity-based redistribution.

Parameter	Dimension	Explanation
$C_{G,P1}$	[1]	EC-incurred grid costs allocated to proximity-area 1
$C_{G,P1HH}$	[HH]	EC-incurred grid costs allocated to each household of proximity area 1
$C_{G,P2}$	[1]	EC-incurred grid costs allocated to proximity area 2
$C_{G,P2HH}$	[HH]	EC-incurred grid costs allocated to each household of proximity area 2
$C_{G,10t}$	[HH]	annual EC-incurred grid costs per HH
GR_{P1}	[T]	generation not used within proximity area 1
GR_{P2}	[T]	generation not used within proximity area 2
GU	[T, HH]	generation used within the EC per time-step and HH
GU_{P1}	[T]	generation used within the EC per time-step within proximity area 1
GU_{P2}	[T]	generation used within the EC per time-step within proximity area 2
LC	[T, HH]	load covered within the EC per time-step and HH
LC_{P1}	[T]	load covered within the EC per time-step within proximity area 1
LC_{P1HH}	[T, HH]	load covered within the EC per time-step and per household within proximity area 1
LC_{P2}	[T]	load covered within the EC per time-step within proximity area 2
LC_{P2HH}	[T, HH]	load covered within the EC per time-step and per household within proximity area 2
LR_{P1}	[T]	residual load of proximity area 1 that has not been covered by generation from proximity area 1
LR_{P2}	[T]	residual load of proximity area 2 that has not been covered by generation from proximity area 1
P	[HH]	array with information to which proximity area the different HHs belong
s_{cross_P1}	[1]	share of proximity area 1 on total cross-proximity-area exchanges
s_{cross_P2}	[1]	share of proximity area 2 on total cross-proximity-area exchanges
s_{LC_P1HH}	[HH]	share of proximity area 1 HHs' load covered within the EC on the total load covered within the EC of all HHs in proximity area 1
s_{LC_P2HH}	[HH]	share of proximity area 2 HHs' load covered within the EC on the total load covered within the EC of all HHs in proximity area 2

Eq. (6) and then further used to calculate the additional costs each non-contributing household has to bear ($C_{G2S_HHnPV}[h]$).

$$s_{LC_HHnPV}[h] = \frac{\sum_{t \in T} LC_{HHnPV}[t, h]}{\sum_{t \in T} \sum_{h \in HH} LC_{HHnPV}[t, h]} \quad (6)$$

$$C_{G2S_HHnPV}[h] = s_{LC_HHnPV}[h] \cdot \sum_{h \in HH} C_{G_EC_HHwPV}[h]$$

Lastly, the EC-incurred grid costs of all households need to be newly determined, meaning that the original EC-incurred grid costs (before redistribution) of...

- contributing households are set to zero
- non-contributing households increase by $C_{G2S_HHnPV}[h]$.

3.3.3. Redistribution based on geographical location

It can reasonably be assumed that it is advantageous for the grid if the sink of energy is close to its source. Or, in other words, the more local energy is consumed, the less the grid is burdened. Thus, in this study two 'proximity areas' are defined within the EC of investigation, each encompassing a number of households that are assumed to be located in immediate proximity:

- Proximity area 1 encompasses households 2, 8, 9, 10, 11, 12
- Proximity area 2 encompasses households 1, 3, 4, 5, 6, 7

The EC-incurred grid costs of all participants shall be redistributed based on the virtual amounts of energy transferred between the two proximity areas. It is assumed that no grid charges apply for amounts of energy exchanged between households in the same proximity area. This implies that the total EC-incurred grid charges are allocated to those amounts of energy that are transferred between participants of different proximity areas. In other words this means that if one proximity area 'imports' higher shares of energy compared to the other proximity area, it will have to bear higher shares of the totally imposed EC-incurred grid costs. The proximity-based redistribution algorithm is explained in detail in the following; the nomenclature is provided in Table 4.

The algorithm, equivalent to the ones before, requires information regarding the proximity of the individual households. This information is provided using the array $P = [2, 1, 2, 2, 2, 2, 1, 1, 1, 1, 1]$, where households located in proximity area one are marked with a '1', and households within proximity area two are marked with a '2'. Depending on to which proximity area a household belongs, the total generation used within the EC and the total load covered within the EC within proximity-areas one and two can be determined (see Eq. (7)), by

extracting respective data out of $GU[t, h]$ – generation used within the EC per time-step and household, and $LC[t, h]$ – load covered within the EC per time-step and household.

for h in range(HH):

if $P[h] == 1$:

$$GU_{P1}[:, h] = GU_{P1}[:, h] + GU[:, h]$$

$$LC_{P1}[:, h] = LC_{P1}[:, h] + LC[:, h]$$

(7)

else if $P[h] == 2$:

$$GU_{P2}[:, h] = GU_{P2}[:, h] + GU[:, h]$$

$$LC_{P2}[:, h] = LC_{P2}[:, h] + LC[:, h]$$

end

In other words, $GU_{P1}[t]$ represents (per time-step) the total amount of PV generation provided to the whole EC by households of proximity-area one. Similarly, $LC_{P1}[t]$ represents the total amount of load covered per time-step for households belonging to proximity area 1. Based thereon, it needs to be distinguished between two scenarios, described in Eq. (8). In case $GU_{P1}[t]$ is larger than or equal to $LC_{P1}[t]$ it means that the load of proximity-area-one households covered by the EC is covered solely by energy that also has been generated within proximity-area one. This implies, that the residual $GR_{P1}[t]$ would then be used within proximity-area two. In case $GU_{P1}[t]$ is smaller than $LC_{P1}[t]$, it means that $LC_{P1}[t]$ cannot be fully covered just by energy generated from households of proximity area one. Instead, there remains a certain amount of residual load $LR_{P1}[t]$ that would be covered from generation of proximity area two. The same procedure is replied for the viewpoint of proximity area two in Eq. (9).

for t in range(T):

if $GU_{P1}[t] \geq LC_{P1}[t]$:

$$GR_{P1}[t] = GU_{P1}[t] - LC_{P1}[t]$$

else if $GU_{P1}[t] < LC_{P1}[t]$:

$$LR_{P1}[t] = LC_{P1}[t] - GU_{P1}[t]$$

end

for t in range(T):

if $GU_{P2}[t] \geq LC_{P2}[t]$:

$$GR_{P2}[t] = GU_{P2}[t] - LC_{P2}[t]$$

else if $GU_{P2}[t] < LC_{P2}[t]$:

$$LR_{P2}[t] = LC_{P2}[t] - GU_{P2}[t]$$

(9)

end

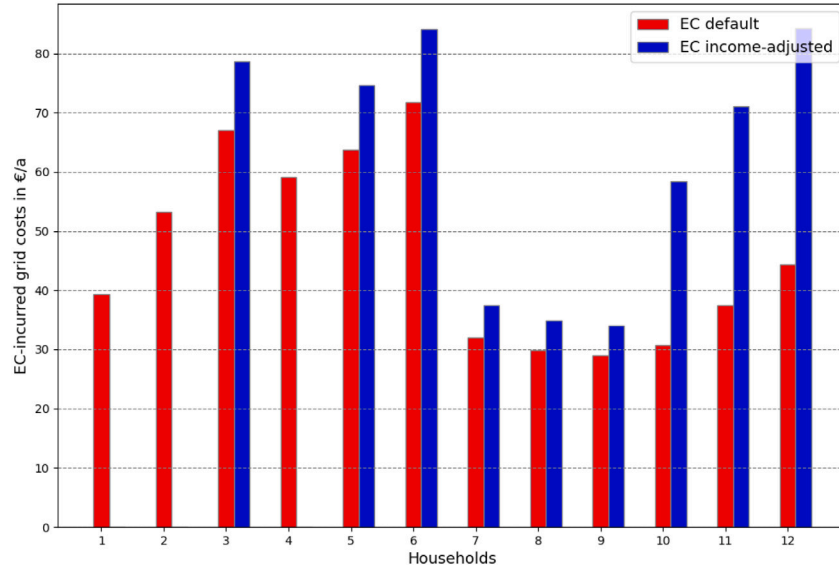


Fig. 2. EC-incurred grid costs per household in the default case (as distributed by the DSO) in comparison to income-based redistribution.

For better understanding, the process above (Eqs. (8) and (9)) imply that $GR_{P1}[t]$ equals $LR_{P2}[t]$, and $GR_{P2}[t]$ equals $LR_{P1}[t]$.

In this case of redistribution based on the proximity of households it is assumed that amounts of energy used within the borders of each proximity area are not burdened with grid costs, while cross-proximity-area energy exchanges are burdened with higher grid costs (since in total, the whole amount of originally arising grid costs need to be covered). Therefore, in Eq. (10), the shares of energy exchanged in total between the two proximity areas are determined. Based thereon, the total EC-incurred grid costs are split between proximity areas one and two generally (not yet distinguished per household).

$$\begin{aligned}
 s_{cross_P1} &= \frac{\sum_{t \in T} GR_{P2}[t]}{\sum_{t \in T} (GR_{P1}[t] + GR_{P2}[t])} \\
 s_{cross_P2} &= \frac{\sum_{t \in T} GR_{P1}[t]}{\sum_{t \in T} (GR_{P1}[t] + GR_{P2}[t])} \\
 C_{G_P1} &= s_{cross_P1} \cdot \sum_{h \in HH} C_{G_tot}[h] \\
 C_{G_P2} &= s_{cross_P2} \cdot \sum_{h \in HH} C_{G_tot}[h]
 \end{aligned} \quad (10)$$

The next step is then to distribute the EC-incurred grid costs of both proximity areas to the individual households within the respective area. To do so, however, it is necessary to determine another criterion for the distribution of costs per household. For that, similarly to the contribution-based algorithm, I use the share of load covered within the EC for the individual households in each proximity area, which is an indicator of how much they benefit from participating in the EC (Eqs. (11) + (12)). However, a prerequisite for this calculation is to first extract information regarding the loads covered per household per proximity area out of $LC[t, h]$ – representing the load covered for each household independent of the proximity area.

for h in range(HH):

if $P[h] == 1$

$LC_{P1HH}[:, h] = LC[:, h]$

else if $P[h] == 2$

$LC_{P2HH}[:, h] = LC[:, h]$

end

$$\begin{aligned}
 s_{LC_P1HH}[h] &= \frac{\sum_{t \in T} LC_{P1HH}[t, h]}{\sum_{t \in T} LC_{P1}[t]} \\
 s_{LC_P2HH}[h] &= \frac{\sum_{t \in T} LC_{P2HH}[t, h]}{\sum_{t \in T} LC_{P2}[t]}
 \end{aligned} \quad (12)$$

In a last step, the EC-incurred grid costs are then finally redistributed to the individual households of the two proximity areas (Eq. (13)).

$$\begin{aligned}
 C_{G_P1HH}[h] &= s_{LC_P1HH}[h] \cdot C_{G_P1} \\
 C_{G_P2HH}[h] &= s_{LC_P2HH}[h] \cdot C_{G_P2}
 \end{aligned} \quad (13)$$

3.3.4. Testing/validating the redistribution mechanisms

The redistribution mechanisms have been thoroughly tested step-by-step, ensuring that they work as expected. Tests encompass, among others, that the sum of redistributed costs equals the original EC-incurred grid costs (as imposed by the DSO). This is of significant importance because costs imposed by the DSO cannot be altered, they just can be redistributed between EC participants. Moreover, the proposed redistribution mechanisms' correct functioning can be proven by looking at the impacts onto different households, depending on the objectives (income-based, contribution-based or proximity-based redistribution).

4. Results

This section aims to present results of the cost-wise implications on individual EC participants for (i) income-based redistribution in Section 4.1, (ii) contribution-based redistribution in Section 4.2, and (iii) proximity-based redistribution in Section 4.3. Section 4.4 concludes the result section by putting all results in direct relation.

4.1. Income-based redistribution

Fig. 2 shows the results achieved by applying the income-based redistribution mechanism to support low-income households. EC-incurred and redistributed grid costs per participant are set in comparison to the default grid costs incurred within the EC (as distributed originally by the DSO). It becomes obvious that the additional financial burden of medium-income households ranges between 5€ per year for household 9, and 12.4€ for household 6. In contrast, the additional burden for households classified as high-income is significantly higher and ranges between 27.6€ (household 10) and 39.9€ (household 12). When seen as pure monetary values these numbers do not seem significant for high-income households, however, it needs to be noted that EC-incurred grid costs of households 10, 11 and 12 are almost doubled when applying income-based redistribution (see Fig. 2). Concluding, the number of medium- and high-income households in an EC (and thus the EC set-up) is decisive for their relative additional financial burden.

Table 5

Comparison of total annual costs: (i) in case no EC would be set up, (ii) in case an EC is established, but no redistribution is applied, (iii) in case an EC is set up and income-based redistribution is applied to EC-incurred grid costs. For situations (ii) and (iii) the percentage of total cost reduction in comparison to the default situation (i) is also provided; those percentages marked in green are the ones that improve based on redistributing EC-incurred grid costs (in other words: larger cost savings in situation (iii) compared to situation (ii)). Percentages marked in red indicate the opposite. For the colour scheme, please refer to the online version of the article.

	Total annual costs in €			
	(i) Without EC	(ii) With EC	(iii) With EC and Redistribu- tion	Economic via- bility
HH1	1137.6	1027.3 (-9.7%)	988.1 (-13.1%)	✓
HH2	1302.9	1153.3 (-11.5%)	1100.0 (-15.6%)	✓
HH3	1455.5	1267.3 (-12.9%)	1278.8 (-12.1%)	✓
HH4	1526.6	1360.5 (-10.9%)	1301.3 (-14.8%)	✓
HH5	1583.5	1404.7 (-11.3%)	1415.7 (-10.6%)	✓
HH6	1869.9	1668.4 (-10.8%)	1680.8 (-10.1%)	✓
HH7	836.9	747.1 (-10.7%)	752.6 (-10.1%)	✓
HH8	1241.8	1158.2 (-6.7%)	1163.3 (-6.3%)	✓
HH9	1043.5	962.1 (-7.8%)	967.1 (-7.3%)	✓
HH10	1228.9	1142.4 (-7.0%)	1170.1 (-4.8%)	✓
HH11	1199.0	1093.9 (-8.8%)	1127.5 (-6.0%)	✓
HH12	1435.9	1311.2 (-8.7%)	1351.1 (-5.9%)	✓

Since households of medium and high income face additional financial burden for the benefit of their low-income peers, it needs to be determined, whether it is still economically viable to participate in the EC. Therefore, the total costs incurred (not solely the EC-incurred grid costs) need to be examined. Table 5 therefore compares the total costs per participant (i) in case no EC would be set up (and thus each household would be on its own), (ii) in case an EC is established, but no redistribution is applied, (iii) in case an EC is set up and income-based redistribution is applied to EC-incurred grid costs.

4.2. Contribution-based redistribution

Fig. 3 juxtaposes default EC-incurred grid costs as imposed by the DSO with EC-incurred grid costs after contribution-based redistribution. What becomes obvious here is that households 7–12 (having a PV system) already have significantly lower default grid costs for EC energy exchanges due to two reasons: Firstly, compared to households 1–6 they purchase less energy from the EC, since significant parts of their load have already been covered by self-consumption. And, for those amounts of energy that households 7–12 sell to their peers, only grid loss charges need to be paid. This means that contribution-based redistribution supports those households, with already smaller amounts of grid charges compared to others. Additional payments for households 1–6 range between 22.5€ (household 1) and 41.2€ (household 6). This means that their EC-incurred grid costs are increased by more than one-third, as Fig. 3 clearly shows.

Table 6 compares the total annual costs per participant (i) in case no EC would be set up, (ii) in case an EC is established, but no redistribution is applied, (iii) in case an EC is set up and redistribution is applied to EC-incurred grid costs based on participants' contribution. Moreover, Table 6 shows that despite contribution-based redistribution increasing the financial burden of households 1–6, economic viability of EC participation is still given.

4.3. Proximity-based redistribution

The case of proximity-based redistribution is specifically interesting for the fictitious EC assumed in this study, because most households owning a PV system are located in the same proximity area (see Fig. 1). Proximity area one (consisting of households 2, 8, 9, 10, 11 and 12) has in total five PV systems at disposal; only household 2 is a sole consumer. In proximity area two, in contrast, there is only one prosumer (participant 7), while the remaining five households are sole consumers. This implies significant amounts of virtual energy flows from proximity area one to proximity area two. In contrast, amounts of

EC generation purchased from proximity area one households originate almost entirely within the same proximity area. And since energy exchanges within the same proximity area are free of grid charges as per the redistribution mechanism, this means that proximity area one households face EC-incurred grid costs close to zero (thus not visible in Fig. 4). In the case of proximity area two, with only one prosumer, large amounts of generation have to be imported from proximity area one. This means that basically all EC-incurred grid costs need to be borne by proximity area two households according to the redistribution algorithm. The range of additional financial burden lies between 29.8€ in case of HH1, and 54.3€ in case of HH6. As Fig. 4 shows, the consumer-households of proximity area two (households 1, 3, 4, 5, 6) record significantly higher EC-incurred grid costs compared to the default setting. Out of proximity area two, only household 7, being a prosumer, records lower EC-incurred grid costs after redistribution due to the set-up of the redistribution algorithm, distributing EC-incurred grid costs based on the shares of the households' load covered within the respective proximity area within the EC. However, despite proximity-based redistribution of EC-incurred grid costs, economic viability of EC participation is still given for all participants, as is confirmed by Table 7.

4.4. Synopsis

To conclude the results, the percentage increase in EC-incurred grid costs for all three redistribution mechanisms is depicted in Fig. 5 for each participating household. With regard to the results of redistributed EC-incurred grid costs in Figs. 2, 3, and 4, the percentages in Fig. 5 might seem surprising at first sight. I would like to explain for the case of contribution-based redistribution: When looking at the case of contribution-adjusted redistribution (the yellow bars in Fig. 5) it can be observed that households 1–6 record equal increases in their EC-incurred grid costs, up to 157%. This equal increase can be explained by the fact that EC-incurred grid costs are redistributed based on the shares of load households 1–6 were able to cover with EC electricity. And since amounts of energy used and grid costs correlate naturally, this leads then to a redistribution of costs that are different in actual numbers (in €), but equal with regard to each household's cost-increase in percent. Now again generally spoken, it can be said that identical increases of EC-incurred grid costs in percent for households in the same situation (e.g. without PV system regarding contribution, medium- or high-income households regarding the financial situation) holds a certain kind of fairness in itself.

Depending on the redistribution mechanism of EC-incurred grid costs, total annual costs of individual participants are influenced. The

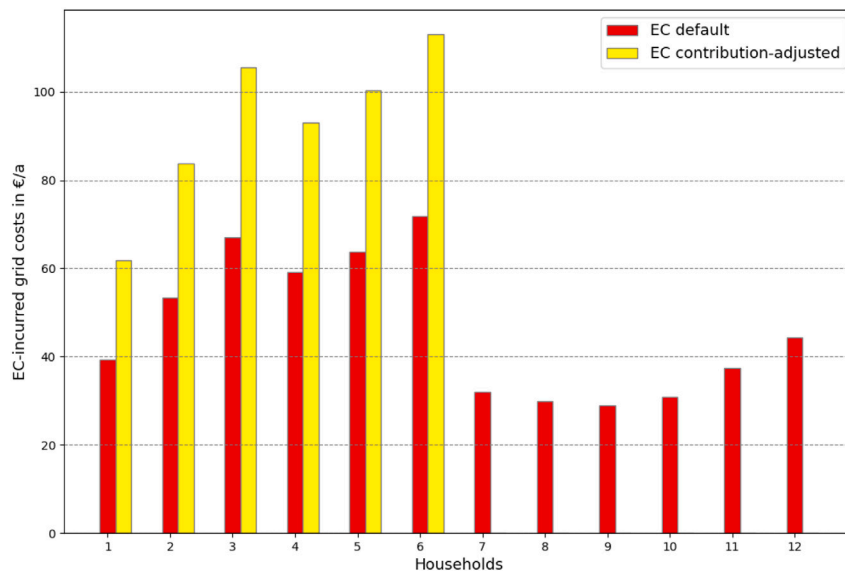


Fig. 3. EC-incurred grid costs per household in the default case of EC establishment in comparison to contribution-based redistribution.

Table 6

Comparison of total annual costs: (i) in case no EC would be set up, (ii) in case an EC is established, but no redistribution is applied, (iii) in case an EC is set up and contribution-based redistribution is applied to EC-incurred grid costs. For situations (ii) and (iii) the percentage of total cost reduction in comparison to the default situation (i) is also provided; those percentages marked in green are the ones that improve based on redistributing EC-incurred grid costs (in other words: larger cost savings in situation (iii) compared to situation (ii)). Percentages marked in red indicate the opposite. For the colour scheme, please refer to the online version of the article.

	Total annual costs in €			Economic viability
	(i) Without EC	(ii) With EC	(iii) With EC and Redistribution	
HH1	1137.6	1027.3 (-9.7%)	1049.9 (-7.7%)	✓
HH2	1302.9	1153.3 (-11.5%)	1183.8 (-9.1%)	✓
HH3	1455.5	1267.3 (-12.9%)	1305.7 (-10.3%)	✓
HH4	1526.6	1360.5 (-10.9%)	1394.4 (-8.7%)	✓
HH5	1583.5	1404.7 (-11.3%)	1441.2 (-9.0%)	✓
HH6	1869.9	1668.4 (-10.8%)	1709.6 (-8.6%)	✓
HH7	836.9	747.1 (-10.7%)	715.1 (-14.6%)	✓
HH8	1241.8	1158.2 (-6.7%)	1128.4 (-9.1%)	✓
HH9	1043.5	962.1 (-7.8%)	933.2 (-10.6%)	✓
HH10	1228.9	1142.4 (-7.0%)	1111.7 (-9.5%)	✓
HH11	1199.0	1093.9 (-8.8%)	1056.5 (-11.9%)	✓
HH12	1435.9	1311.2 (-8.7%)	1266.9 (-11.8%)	✓

Table 7

Comparison of total annual costs: (i) in case no EC would be set up, (ii) in case an EC is established, but no redistribution is applied, (iii) in case an EC is set up and proximity-based redistribution is applied to EC-incurred grid costs. For situations (ii) and (iii) the percentage of total cost reduction in comparison to the default situation (i) is also provided; those percentages marked in green are the ones that improve based on redistributing EC-incurred grid costs (in other words: larger cost savings in situation (iii) compared to situation (ii)). Percentages marked in red indicate the opposite. For the colour scheme, please refer to the online version of the article.

	Total annual costs in €			Economic viability
	(i) Without EC	(ii) With EC	(iii) With EC and Redistribution	
HH1	1137.6	1027.3 (-9.7%)	1057.1 (-7.1%)	✓
HH2	1302.9	1153.3 (-11.5%)	1100.1 (-15.6%)	✓
HH3	1455.5	1267.3 (-12.9%)	1318.0 (-9.4%)	✓
HH4	1526.6	1360.5 (-10.9%)	1405.3 (-7.9%)	✓
HH5	1583.5	1404.7 (-11.3%)	1452.9 (-8.2%)	✓
HH6	1869.9	1668.4 (-10.8%)	1722.7 (-7.9%)	✓
HH7	836.9	747.1 (-10.7%)	743.4 (-11.2%)	✓
HH8	1241.8	1158.2 (-6.7%)	1128.4 (-9.1%)	✓
HH9	1043.5	962.1 (-7.8%)	933.2 (-10.6%)	✓
HH10	1228.9	1142.4 (-7.0%)	1111.7 (-9.5%)	✓
HH11	1199.0	1093.9 (-8.8%)	1056.5 (-11.9%)	✓
HH12	1435.9	1311.2 (-8.7%)	1266.9 (-11.8%)	✓

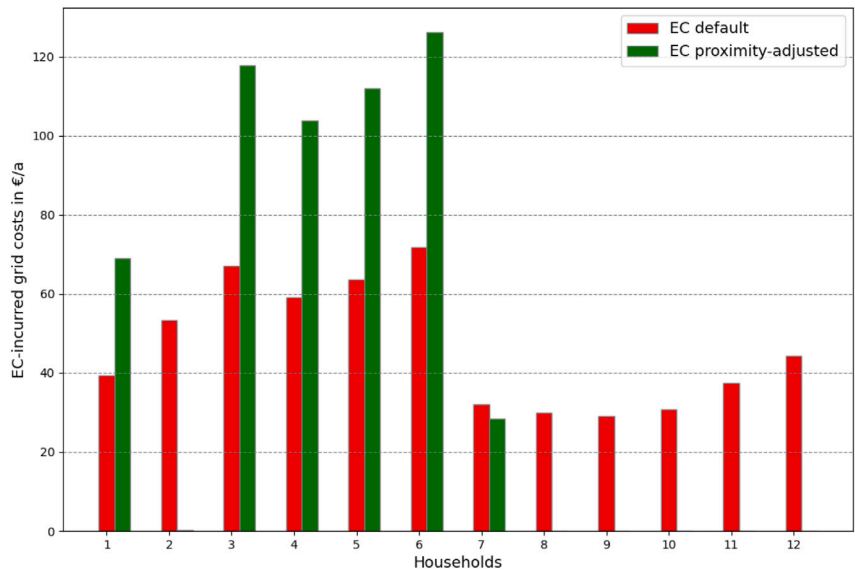


Fig. 4. EC-incurred grid costs per household in the default case of EC establishment in comparison to proximity-based redistribution.

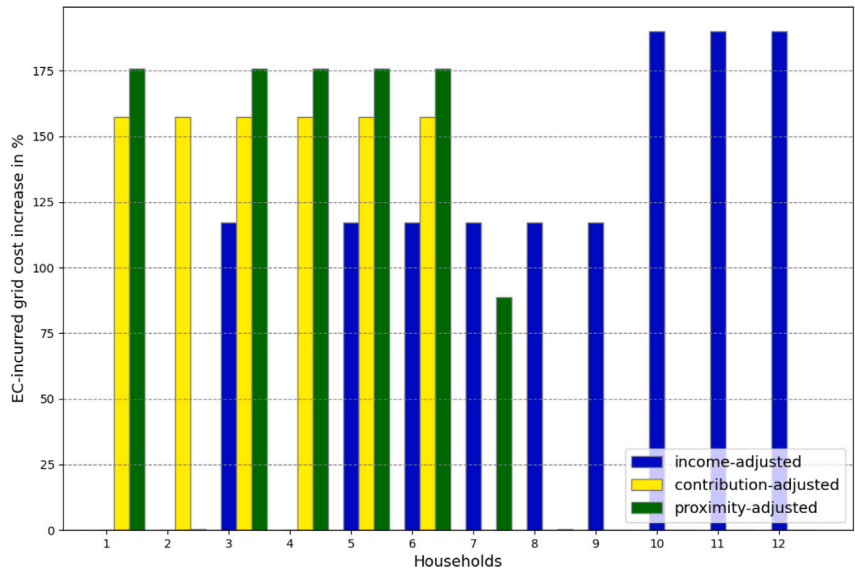


Fig. 5. Increase of EC-incurred grid costs in percent after redistribution with respect to the EC-incurred grid costs without applying any redistribution mechanism.

negative or positive influence on each household's total annual costs is lastly depicted indicatively in Table 8. For a variation of total annual electricity costs in the range of $0\% < x < 2\%$ symbols of $\uparrow$ and $\downarrow$ are used; cost variance in the range of $2\% < x < 4\%$ is depicted with $\uparrow\uparrow$ and $\downarrow\downarrow$, and cost variance $> 4\%$ is marked with $\uparrow\uparrow\uparrow$ and $\downarrow\downarrow\downarrow$. For all cases accounts: A positive influence of redistribution mechanisms of EC-incurred grid costs on the total annual electricity costs is indicated with uparrow(s), while a negative influence is indicated with downarrow(s).

5. Discussion

Here, aspects of fairness for each of the proposed redistribution mechanisms are discussed in Sections 5.1, 5.2, 5.3, before concluding thoughts with regard to fairness are presented in Section 5.4. Moreover, principles of grid tariff design are applied to the proposed redistribution mechanisms to check for consistency (even though the proposed redistribution mechanisms are not a grid tariff design, but can be applied to grid costs based upon different grid tariff designs) in Section 5.5.

Table 8
Variance of total annual electricity costs for each household in case of applying the three different redistribution mechanisms to EC-incurred grid costs. (Based upon percentages provided in Tables 5, 6 and 7.)

	Variance of total annual costs		
	Income-based	Contribution-based	Proximity-based
HH1	↑↑	↓↓	↓↓
HH2	↑↑↑	↓↓	↑↑↑
HH3	↓	↓↓	↓↓
HH4	↑↑	↓↓	↓↓
HH5	↓	↓↓	↓↓
HH6	↓	↓↓	↓↓
HH7	↓	↑↑	↑
HH8	↓	↑↑	↑↑
HH9	↓	↑↑	↑↑
HH10	↓↓	↑↑	↑↑
HH11	↓↓	↑↑	↑↑
HH12	↓↓	↑↑	↑↑

5.1. Fairness of income-based redistribution

One major point in favour of income-based redistribution is that it opens up a possibility to better include households with limited financial means by providing additional financial alleviations in addition to those benefits already achieved by EC participation *per se*. This way of redistribution also tries to tackle claims posed by the European Directives, namely to alleviate energy poverty and include vulnerable customers.

In this study, it is assumed that one-third of low-income households' EC-incurred grid costs are borne by medium-income households, the remaining two-thirds by high-income households. This is based on the logic of distinguishing between three income categories. However, two questions need to be asked at this point. Firstly, how many low-income households are 'affordable' by an EC in general? It would need to be determined how many low-income households could be included in an EC until the financial situation of at least one medium- or high-income household worsens such that EC participation is not beneficial any longer. Secondly, what if the number of medium- and high-income households is very unevenly distributed, e.g. multiple medium-income households and only one high-income household? According to the redistribution mechanism, this would mean that this one high-income household would have to bear the two-thirds of low-income households' EC-incurred grid costs alone. Thus, from this viewpoint it can be concluded that the practical application of such redistribution mechanism needs to be carefully considered based on the set-up of each EC individually.

Besides, applying this form of redistribution depends significantly on the social-mindedness of all participants, as well as on whether participants actually openly disclose their financial situation. Full disclosure of income could be a hurdle especially for low- as well as for high-income households. And, if participants would need to allocate themselves to one of the three income-levels (in order to avoid full disclosure of income), there is the danger of strategic behaviour, benefiting some, while disadvantaging others. In this context, there would be the need to clearly draw a line between medium- and low-income households, as well as between high- and medium-income households. However, defining at which point a household is considered low-income is the most important question to be raised. For being able to define what a low-income household actually is, a clear definition of 'energy poverty' would be of help. While research has already tried to tackle this definition, no unified solution could be proposed, since energy poverty depends on multiple factors, and additionally differs from country to country (BMS, 2016; Statistik Austria, 2022). Concluding, when implementing an income-based mechanism to redistribute EC-incurred grid costs, fairness of such approach needs to be thought through on multiple levels.

5.2. Fairness of contribution-based redistribution

In comparison to income-based redistribution of EC-incurred grid costs, applying the contribution-based mechanism follows a more obvious rational principle that does not require the households to disclose private information. It is obvious, which households are equipped with a PV system, and which are not. Moreover, it is a fact that an EC could not exist without energy generation, wherefore it is comprehensible why participants with a PV system are financially advantaged by the redistribution algorithm. However, the relation of both participant types (participants with and without PV system) needs to be in proportion to make the contribution-based redistribution algorithm practically applicable (such that also non-contributing households still benefit from EC participation).

Typically, PV system implementation and the respective grid feed-in of generated electricity has an impact on the distribution grid. Increased feed-in into the grid increases the frequency, which can lead to grid stability issues in case of too little consumption at the same time.

Thus, whether installation of PV systems alleviates or increases the grid burden also depends on the individual situation (e.g. distribution grid structure, characteristics of connected grid users). This would mean that from a grid-perspective, reduced grid costs for households equipped with a PV system would only be reasonable, if PV systems reduced the grid burden. However, in this study the perspective of the EC itself is relevant, wherefore the contribution of households (equipped with a PV system) is rewarded with the omission of EC-incurred grid costs.

5.3. Fairness of proximity-based redistribution

After all, proximity-based redistribution of EC-incurred grid costs is the only option that truly and unanimously also complies with grid requirements, namely that decentralised generation shall be consumed as local as possible. In this study, households are divided into belonging to two different proximity areas. With establishing EC-incurred grid cost redistribution such that local energy exchanges are financially privileged, local consumption would be incentivised when implementing this algorithm in practice. Thus, from the viewpoint of the DSO, this redistribution mechanism would make the most sense. Also, this kind of redistribution would be understandable from a rational (or physical) point of view by the EC participants. Furthermore, no personal information of the individual participating households would need to be disclosed.

However, the aspect of fairness of proximity-based redistribution is still questionable, since participants cannot 'choose' to just move to the proximity area where the most PV generation is available. Moreover, from a practical-application point of view, problems might arise if the majority of households is located in the more favourable proximity area with large amounts of PV generation, while only a small number of households is located in the less favourable proximity area. The latter would be required to 'import' most EC energy from the more favourable proximity area and thus would be burdened with the majority of EC-incurred grid costs. So, also in this case of redistribution, the practical applicability needs to be validated based on the individual set-up of the EC.

5.4. Concluding thoughts on fairness aspects

Fairness is a subjective perception, depending on a person's background, their character, education, and else. Therefore, it is highly likely that like-minded people sharing certain values join together for an EC. The three redistribution mechanisms proposed in this study built upon different possible fairness perceptions for EC-incurred grid cost distribution. However, all three redistribution mechanisms have in common that with their application, certain EC participants benefit, while others have to bear additional costs. The question that needs to be raised is what motivates participants to agree to such cost-redistribution in case they are not the ones who benefit financially. The answer to that is simple and complex at the same time: It is all about personal values. Income-based redistribution of EC-incurred grid costs could be viewed as a form of donation to the more vulnerable. Contribution-based redistribution of EC-incurred grid costs can be viewed similar to the case, when end-users agree to voluntarily pay more for being supplied with green electricity from a conventional supplier. And, proximity-based redistribution might seem logical to EC participants since it is a fact that using energy close to its place of origin is more efficient. The proposed redistribution mechanisms for EC-incurred grid costs are three examples of how such redistribution could be realised, following three possible objectives/values of an EC and their participants.

Table 9
Principles of grid tariff design (Reneses and Ortega, 2014, Vaughan et al., 2023) applied to the novel redistribution mechanisms.

	income-based	contribution-based	proximity-based
sustainability	✓: costs incurred from the DSO are fully covered; proximity-based redistribution could lead to increased sustainability in the long term, since local energy trades are financially advantageous.		
equity	✓: aiding the less-fortunate	equity, per definition leading to social justice, is not given	equity, per definition leading to social justice, is not given
non-discriminatory access	✓: given, since the less-fortunate receive additional financial advantages	depends: if EC participation still profitable after redistribution – yes; otherwise – no	depends: if EC participation still profitable after redistribution – yes; otherwise – no
economic efficiency	given for members of an EC with a balanced composition of participants – however, needs to be evaluated per individual case; always given for the DSO since incurred costs are only redistributed		
transparency	all redistribution mechanisms follow transparent criteria		
stability	✓	✓	✓
consistency with larger regulat. framework	✓	basically yes; however full compliance with EU regulations ‘... accessible to all consumers, including those in low-income or vulnerable households;’ (REDII, Article 22 (4f)) is questionable	
additivity of cost elements	✓	✓	✓
privacy	no, since participants would need to fully disclose their financial situation	✓: PV systems are obvious without needing to disclose private information	✓: proximity is obvious without need to disclose private information
implementability	general applicability is always given; whether the redistribution mechanisms are sensible implementable (with regard to EC set-up and participant structure) needs to be evaluated individually		

5.5. Principles of grid tariff design applied to redistribution mechanisms

The proposed redistribution mechanisms are not grid tariffs, but can be applied to grid costs based upon different grid tariff designs. Thus, redistribution mechanisms are neither implemented by the DSO, nor do they directly affect the DSO. The implementation happens within the EC, based on the joint agreement of all participants. The independence of redistributing EC-incurred grid costs thus implies that the principles or objectives of the different redistribution mechanisms (income-, contribution- or proximity-based redistribution) are not required to comply with DSOs’ objectives (stable grid operation, reducing peak load etc.), but solely aim to realise subjective perceptions of distributional fairness within the EC. However, it would certainly be beneficial, if subjective perceptions of fairness would also be in line with DSOs’ objectives. With regard to the here proposed redistribution mechanisms, only proximity-based redistribution can be considered to be in line with DSOs’ objectives, since close proximity of energy source and sink are beneficial with regard to keeping the grid burden possibly low. Whether contribution-based redistribution is compatible with DSOs’ objectives depends significantly upon each individual situation. In case additional benefits for PV system owners (in the form of alleviating the burden of EC-incurred grid costs through redistribution) motivate additional PV installations in grid sections with relatively high loads and comparably low generation (which would grid-wise lead to tendentially lower grid frequency), compliance with DSO objectives would be given. The third mechanism which is based on redistributing EC-incurred grid costs based on households’ income levels cannot be connected to any objective relevant to a DSO.

However, as described in the course of the literature review, grid tariff design usually follows a number of principles. In the following it shall be evaluated whether the newly developed redistribution mechanisms (not being a grid tariff, but certain similarities regarding principles might be given) comply with conventional grid tariff design criteria. Table 9 depicts tariff principles identified by Reneses and Ortega (2014) and Vaughan et al. (2023), and assess validity for (i) income-based redistribution, (ii) contribution-based redistribution, and (iii) proximity-based redistribution.

6. Conclusion

The developed redistribution mechanisms (income-, contribution- and proximity-based) for EC-incurred grid costs are deemed suitable

for ECs to realise subjective goals regarding distributional fairness and provide EC participants with an additional degree of freedom since grid costs are usually imposed by the DSO. A significant strength of this study is that the proposed redistribution mechanisms are not restricted to only being applicable to EC-incurred grid costs, but can be used for any kind of cost component that needs redistribution. Moreover, the algorithms are developed such that an application to any kind of EC is possible, independent of the types (households, SMEs, etc.) or number of participants. Since the proposed redistribution mechanisms are the first of their kind, they can also be used as inspiration for the establishment of any other redistribution mechanism that might seem fit for EC application. It needs to be mentioned that it would have also been possible to address the issue of fairness (to benefit some participants more than others) through an optimisation of the EC-internal energy price. The decision to develop redistribution algorithms for EC-incurred grid costs instead is twofold. Firstly, ECs are free to choose their EC-internal energy price, while grid costs are simply imposed by the DSO. With redistribution mechanisms for EC-incurred grid costs, ECs are provided with an additional degree of freedom. Secondly, it is not the goal to achieve ‘the best possible economic solution for some participants’ (as using optimisation would suggest), but the goal is rather to achieve an acceptable situation for all participants.

A limitation of this work is that, although being generally applicable to any EC, practical application needs to be critically evaluated situation-specific. For the case of income-based redistribution this would mean to make sure that a sufficient number of medium- and/or high-income households are present in relation to low-income households, such that it can be ensured that their additional financial burden remains manageable (e.g. a community of 10 low-income households and 1 medium-income household would put all the additional financial burden on 1 medium-income household, which would not be a ‘healthy’ precondition for applying income-based redistribution; apart from that, the medium-income household would most probably not agree to income-based redistribution anyway.). For the case of contribution-based redistribution practical applicability needs to be checked by making sure that a sufficient number of sole consumers are present. If, e.g. an EC would consist of prosumers and just one sole consumer, contribution-based redistribution would request for the sole consumer to overtake all EC-incurred grid costs of the prosumers. In the case of proximity-based redistribution it would make sense to ensure that the different proximity areas have a balanced

composition of participants with and without PV systems. Concluding, despite being generally applicable without doubt, a ‘sanity-check’ needs to be conducted for any EC set-up. Thereby, it needs to be emphasised that – up to this point – it is not possible to generalise exactly what an EC should or should not look like in case of applying one of the proposed redistribution mechanisms. The only statement with general validity is that diversity in the EC composition is never wrong and has the highest potential to ensure a ‘healthy’ application of the redistribution mechanisms. This, however, leads to potential future research topics: It would be of significant interest to elaborate on ‘the critical set-up’ of an EC for all three redistribution mechanisms to be sensibly applicable such that economic viability of EC participation is maintained for all EC members. Moreover, other pieces of future research could investigate whether different redistribution mechanisms could be linked – from single-objective to multi-objective redistribution. However, it is highly likely that combining certain redistribution mechanisms leads to cannibalising effects. For example, redistribution based on income-levels would be highly contradictory to contribution-based redistribution because it can be assumed that only medium- or high-income households would be able to afford investing in a PV system, while low-income households could not. In other words, income-based redistribution benefits the poor, while the contribution-based mechanism most likely benefits the wealthier EC participants. However, it would be possible to introduce weighting factors to combine even contradictory or conflicting objectives when it comes to redistribution.

CRediT authorship contribution statement

Bernadette Fina: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Author declaration

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Appendix A. Energy communities throughout Europe

With the enactment of the European Directives RED II [EUR-Lex \(2018\)](#) and EMD [EUR-Lex \(2019\)](#), EU member states have been given a time frame of 1–2 years at maximum to transpose these supranational guidelines into national legislation. The transposition process was approached quite differently by the individual countries; some acted as fore-runners, such as Austria and Italy, while others were unexpectedly reserved towards transposing EC guidelines, such as Germany. However, what all countries have in common is that they interpreted EU guidelines differently, leading to a broad spectrum of transposition designs. Parameters which can be found to be interpreted differently by legislators of different EU countries are for example (a) proximity of REC participants, (b) implementation of energy sharing,

(c) possibility of acting as a grid operator, (d) energy pricing, (e) exemptions/reductions of grid tariffs and levies, (f) offered subsidies, and else. Taking the proximity constraint as an example, Austria defines the proximity of RECs via grid levels, proposing that participants of local RECs need to be connected via the same low-voltage grid feeder, while participants of regional RECs need to be connected by the same medium-voltage grid feeder ([Rechtsinformationssystem des Bundes, 2023a](#)). Similarly, Italian legislators define proximity of RECs such that participants need to be allocated under the same HV-MV substation. In Germany, proximity is bound to the rule that 75% of voting rights of a REC need to be held by natural persons living in a zip code area that is wholly or partially within 50 kilometres ([Bundestag, 2023](#)), and French legislators decide that proximity means residence or location in the same or a bordering *département* ([RESCoop, 2023](#)). Another example for significant transposition differences is how energy sharing is realised. In Austria, DSOs are legally obliged to perform the allocation for the ECs, based on a static or dynamic distribution key. As a result, the EC participants receive energy data (in 15-min resolution on the following day) with information of how much energy has been sold to the EC (in case of owning a PV system), and how much energy has been purchased from the community. In Italy, in contrast, the issue of energy sharing is solved by matching generation and demand of an EC (aggregating all participants’ generation and demand profiles) on an hourly basis, while Germans need (to the author’s current best knowledge) an energy supplier offering an adequate service for energy sharing. These examples shall provide a glimpse into how differently ECs are managed throughout Europe, while a more detailed investigation would be out of this study’s scope.

Appendix B. Fairness being a matter of individual perception

Since this study’s proposed redistribution mechanisms for EC-incurred grid costs are based on stressing the concept that fairness is perceived differently by individuals, it is necessary to go into that in a little more detail. Grid tariff design needs to follow a number of principles, whereas one of these principles is equity ([Reneses and Ortega, 2014](#)). Equity refers to ‘fairness’ and ‘justice’, wherefore it can be concluded that a grid tariff is required to be fair by default. Nevertheless, different grid tariff designs are heatedly discussed and perceived differently by household customers with regard to fairness, as has been proven through surveys conducted by [Trong and Yang \(2023\)](#) and [Neuteleers et al. \(2017\)](#). Not only vary fairness perceptions between household customers, but are also perceived differently by different stakeholders ([Winther and Sundet, 2023](#)). In a broader energy-related context, [Bal et al. \(2023\)](#) argue that ‘*individual-level perception measures are insufficiently developed ..., while they are vital for a successful just energy transition*’. These authors also highlight a range of studies addressing fairness perceptions regarding the energy transition, such as [Steg et al. \(2015\)](#), [Huijts et al. \(2012\)](#), [Perlaviciute and Steg \(2014\)](#), [Gözl and Wedderhoff \(2018\)](#).¹⁸ This shows that a wide range of scientific literature has recognised and acknowledged the fact that fairness is a concept that is subject to individual perceptions, wherefore it is valid for this study to built upon.

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¹⁸ A large variety of other studies in this or other similar directions are provided in the course of [Bal et al. \(2023\)](#).

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